Reset-free syndrome extraction on IBM heavy-hex: replication, pre-registered tests and honest negative results

Research update · 15 October 2026 · Ralf Novikov · code and raw data: github.com/ralfnovikov1-spec/quantum-research · analysis prepared with AI assistance (Claude, Anthropic)

In one paragraph. On IBM's 156-qubit Heron processors, removing mid-circuit resets from heavy-hex syndrome extraction lowered the Z-basis logical error per round by about ×1.7 on ibm_kingston, in four runs, each 2σ-separated. That includes a pre-registered rerun at 10,000 shots. This independently replicates Harper et al. (2025), who showed the effect first on ibm_marrakesh. All 9 hardware jobs were re-fetched from IBM and re-derived with independent code. Two new ideas were tested on hardware under pre-registration: probe-informed decoding **failed**, and leakage-aware decoding **failed** to improve the logical error, although leakage-consistent readout events were clearly detected. A daily automated study is now running, with three claims fixed in advance.

1. What holds, what failed

Claim	Evidence (hardware unless marked)	Status
Removing resets lowers Z-basis logical error per round (d = 3)	×1.70, ×1.66, ×2.07, ×1.72 on ibm_kingston; 2σ-separated; last run pre-registered, 10,000 shots	**SUPPORTED**
Same in the X basis	×1.13–1.42	**SUPPORTED (smaller)**
Same on ibm_marrakesh	×1.07 / ×1.05, inside 2σ	**DIRECTION ONLY**
Break-even (logical outlives best physical qubit)	0.092× (Z), 0.136× (X), pre-registered	**NOT REACHED**
A seconds-long live probe makes the noise model accurate	within ±30% (4/4 arms) vs 1.5–2.1× optimistic from calibration; one day, one chip; decoder caveat (see fix log)	**SUPPORTED, CAVEAT**
Detectors that over-fire sit on qubits whose readout drifted	23 qubit sets: ρ = +0.54 [0.17, 0.80], p = 0.009; T1 drift: no detectable association	**EXPLORATORY → C3 daily**
Probe-informed decoding lowers logical error	6/16 arms better; 0.69% more errors; p = 0.999 (pre-registered D1–D3)	**FAILED**
Leakage visible in analog (IQ) readout	persistent off-axis events, ratio 105, p ≈ 10^{-186}, 12/14 ancillas (pre-registered L0)	**DETECTED**
Leakage-aware decoding lowers logical error	3338 → 3394 errors (+1.7%); persistent-flag variant neutral (post hoc)	**FAILED**
Larger codes help (Λ > 1)	only d = 3 run on hardware	**NOT TESTED**

2. Prior work and what is not new

Reset elimination on heavy-hex was published first by **Harper et al.**, arXiv:2504.07258, *Nature Communications* 17, 5039 (2026): ibm_marrakesh, d = 3, 96–97% per-round fidelity with a shorter circuit. Reset-free extraction was analysed in simulation by Gehér et al. (arXiv:2408.00758). IBM's 2022 heavy-hex experiment is arXiv:2203.07205. Soft-information decoding on IBM hardware:

arXiv:2411.16228. Noise-aware decoding: arXiv:2502.21044, 2406.02700. Leakage detection with analog readout: Varbanov et al., npj QI 2020. This work is an **independent replication** on another chip with IBM's original flag-qubit circuit, plus the tests listed above. Novelty claims rest on targeted searches, not a full review.

3. Hardware runs (all retrievable by job ID)

Date	Job	Chip	What	QPU s
14 Oct	db7r4per…, db7rf1gr…, db7s2dar…, db7rbner…	kingston / marrakesh	reset-free vs baseline runs 1–3 + Marrakesh	25–36 each
14 Oct	db7qkvei…, db7qqker…, db7rid8r…, db7s4t6r…	marrakesh / kingston	DD, placement, probe, break-even	19–47 each
15 Oct	db88jo6r… + db88kdgr…	kingston	live probe + pre-registered rerun (30 circuits × 10k)	99 (main)
15 Oct	db8a5aer…	kingston	daily study, day 1	≈ 35 (incl. probe)
15 Oct	db8clieil…	kingston	pre-registered leakage pilot, kerneled IQ	9

4. Fix log (everything that was corrected)

Report corrections (14 Oct → 15 Oct)

- Eight pre-registered predictions that had failed are now marked FAILS (P3, P4, P6, P10, P13, P15, P16, P19). Some had been shown as "Mixed" or not mentioned.
- The 14 Oct break-even "pass" was obtained after excluding a dead qubit (q146) post hoc. It is now labelled that way and was retested cleanly with a dead-qubit filter applied before selection.
- Dynamical-decoupling table: the arms without DD were saturated (45–50% error after 3 rounds), so the ×1.21–1.27 magnitude was withdrawn. The per-round metric there differed from the rest of the report.
- Added shot counts and sources for every number. The "$\Lambda < 1$ on hardware" claim was withdrawn (never run at $d = 5$).

Independent code and physics reviews

- The re-derivation script merged two DD arm types; fixed. All 8 arms then reproduce, including P6 = 7.48%.
- Break-even lifetime model: a review concern (2.6× error) was checked against the exact DD pulse-sequence model. The original fit was within about 5%, and the Z ratio was corrected from 0.096× to 0.092×.
- The probe-model prediction had been decoded with its own noise model instead of the calibration decoder used on the hardware data. This was fixed for all future runs, and the 15 Oct result is caveated.
- Daily pipeline: no same-day resubmission; plan saved before submission; budget guard fails closed; per-day error records; results always committed. The protocol was bumped to v2 before any daily result.

False positives caught by simulation before hardware

- Noise spectroscopy: a "≥ 60% of qubits" rule passed on pure white noise. It was replaced with a sign test plus effect size, giving 0/20 false positives.
- A Ramsey measured only in X gave **10/10 false positives** under static frequency offsets. It is now X+Y magnitude with per-qubit readout correction, giving 0/20 with confounds (offsets, ZZ, asymmetric readout).
- Leakage decoding: readout-only erasure *hurts* once leakage spreads to neighbours, so the neighbour-aware decoder was built. False-flag cost was quantified, and a binomial persistence test now gates the flags.

Audit of earlier exploratory notebooks

31 earlier simulation notebooks were audited, fixed and re-run. Several headline effects turned out to be implementation bugs: a reservoir that reset every step; an 8-bit integer overflow; a non-Hermitian Hamiltonian; integrator norm drift; and hard-coded "CONFIRMED" strings. With the bugs fixed, those claims were withdrawn. Correct reproductions of known physics remained, for example the toric-code threshold at 10.38% against about 10.3% in the literature. These notebooks are not published.

5. Running now: pre-registered daily study

A GitHub Action runs every day on IBM hardware, alternating ibm_kingston and ibm_marrakesh. It probes all qubits, runs baseline and reset-free arms, and publishes `daily/RESULTS.md`. Claims fixed

before any data:

- C1: the calibration-fed model is at least 2× less accurate than the probe-fed model on at least 80% of days.
- C2: the reset-free Z gain is 2σ-separated on at least 80% of days.
- C3: detector excess tracks readout drift on at least 60% of days.

No verdict before 20 analysed days.

6. Limitations and next steps

- Distance 3 only. A matching decoder only. Few days per chip. IBM open-plan budget: 600 s of QPU per month.
- The leakage events could not be calibrated (|2⟩ cannot be prepared on Heron), so no leakage *rate* is claimed.
- Next: let the daily study reach 20 days. If more QPU time becomes available, test whether leakage accumulates over many rounds (pre-registered), and run noise spectroscopy on hardware.

All claims are traceable to files in the repository. Simulations are labelled as such and say nothing about hardware.